

# E-Commerce in Agriculture: A Review of Digital Marketplaces for Direct Farm-to-Consumer Trade :

## INTRODUCTION :

Agriculture is very important for the economy of countries especially in India where many people depend on farming for their living. Milk is one of the important food items that people consume. The dairy industry helps the economy a lot by giving farmers a chance to earn money and providing nutrition to consumers.. Even though milk production is so important the traditional milk supply chain still has many problems that affect both farmers and consumers. In the milk distribution system milk goes through many people like local collectors, wholesalers, distributors and retailers before it reaches the consumer. These people often take a share of the profit leaving farmers with very little money for their work. As a result farmers are often not paid enough for their milk. Have little control over the price of milk. Also farmers often have to wait a time to get paid which causes financial problems, for small dairy farmers who need to earn money every day to live. When we buy milk we do not always know where it comes from or how fresh it is. The milk supply chain has a lot of people involved which means the milk has to travel and be stored for more time. This can make the milk less fresh and less good for us. Also we do not always know if the milk is safe and really comes from a dairy farm. New technology is changing the way we buy food from farms. A study by Tajammul et al. In 2025 says that online platforms can help farmers sell directly to us. This means we do not need to rely on many people in, between. These online platforms make it easy for us to talk to farmers and pay a price. They also help farmers make money and give us fresh food. By using these platforms farmers and people who deliver the food and we, the people who buy the food are all connected. This makes everything work better. We get good food.

This project is about creating a platform that connects dairy farmers, delivery partners and consumers directly. The main goal is to make milk distribution simpler by cutting out middlemen and letting farmers sell milk to consumers. Farmers can list their milk products manage orders and get paid instantly. This way they get pay for their work. Consumers can buy milk from nearby farms and know exactly where it comes from and its quality. The platform also helps delivery partners do their job efficiently. It has features like tracking orders in time digital payments and organized delivery. These features make the whole milk supply chain process smoother. This approach makes milk distribution more efficient. It also helps build trust between farmers and consumers by giving information about the milk. In conclusion using technology in farming supply chains can solve many problems with traditional milk distribution. This project aims to help farmers give them prices and provide consumers with fresh, safe and traceable milk. The platform helps build a dairy supply chain that's transparent, efficient and sustainable. It connects farmers, delivery partners and consumers in an fair way. The goal is to make milk distribution better, for everyone involved.

### **CITATION :**

1. Mohd Tajammul, Shivam Singh, Atul Kaintyura, Babita Kumari, and Rajat Tomar, "E-Commerce in Agriculture: A Review of Digital Marketplaces for Direct Farm-to-Consumer Trade," *International Journal of Engineering Research & Technology (IJERT)*, vol. 14, no. 05, May 2025.

### **BIB TEX :**

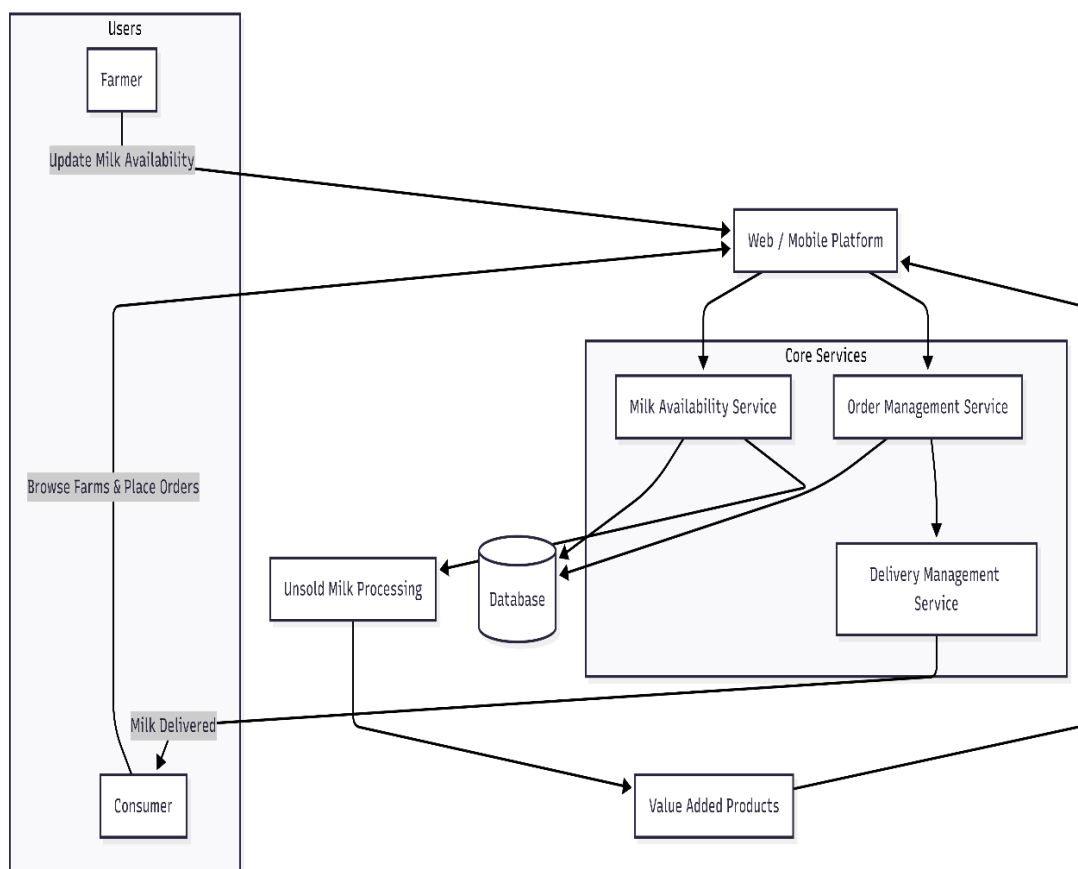
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## 2. METHODOLOGY :

In this section, the design and implementation of the proposed Hyperlocal Direct Milk Supply Platform will be discussed. The proposed system will be used for the direct supply of milk from dairy farmers to end users/consumers without any intermediary.

The proposed methodology has discussed the overall architecture, frontend development, backend development, database schema, APIs, security features, technology stack, and implementation strategy for the proposed system.

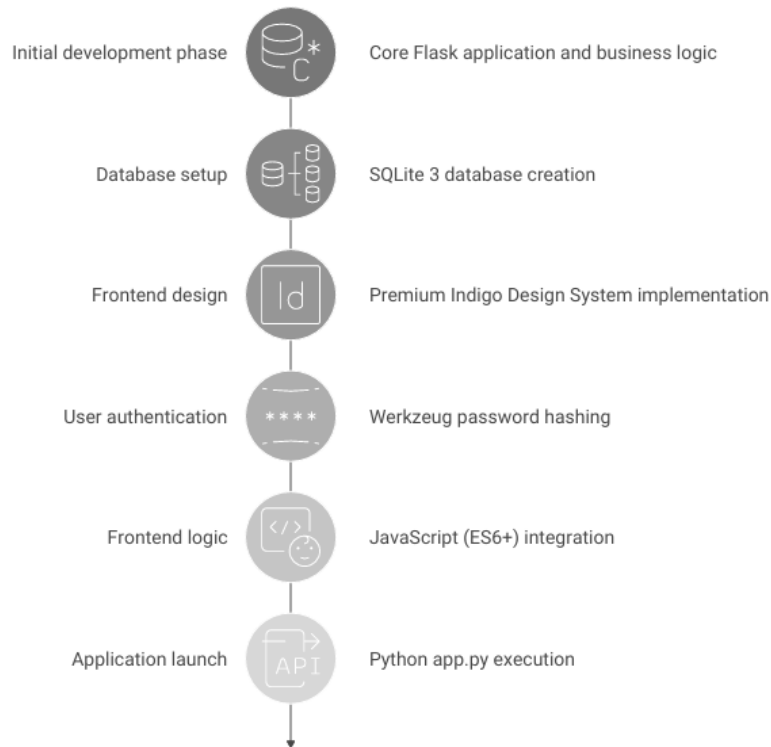
### 2.1. ARCHITECTURE DIAGRAM :



The above diagram shows the architecture diagram for the Hyperlocal Direct Milk Supply Platform. Here, farmers and consumers interact with each other via the web/mobile application. Farmers update their milk availability on a daily basis, and consumers can view the farmers' locations to order milk products. The Milk Availability Service, Order Management Service, and Delivery Management Service are the backend services that process the requests made by the customers. The requests are stored in the central database. The Unsold Milk Processing module is used to convert unsold milk into value-added products like paneer, butter, or ghee to minimize wastage.

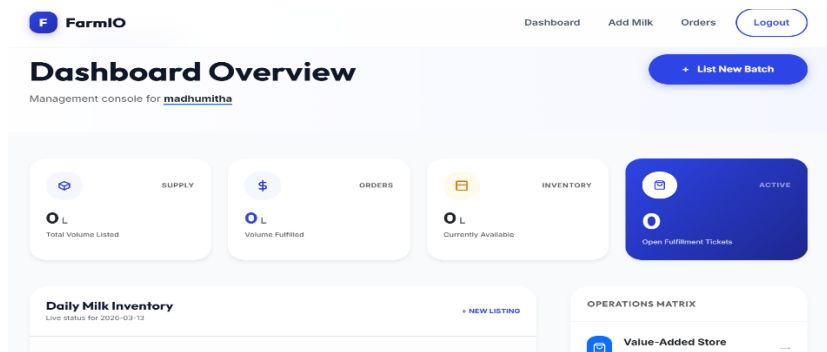
### 2.1.1. TECHNICAL STACK :

#### Farmio's Journey: From Concept to Consumer



The above workflow diagram shows the process of developing the Farmio platform.. The process starts with the Flask application development and database setup using SQLite. Then the frontend interface and user authentication system are designed using the Indigo design system, JavaScript, and password hashing. Finally, the application is deployed by running the Python Flask application (app.py) to enable farmers and consumers to interact with each other.

### 2.2. FRONTEND DESIGN :



In the above image, we can see a representation of the FarmIO farmer dashboard interface where farmers can easily manage their milk supply operations. In the above interface, farmers can easily track their total milk supply, orders fulfilled, current inventory, and active delivery tickets in a card-based view. Farmers can easily list their new milk batches, daily inventory, and value-added products with ease

### **2.3. BACKEND SERVICES :**

The backend server controls the major operations of the system and coordinates the interactions between the frontend interface and the database. It receives user requests, performs operations based on the system's logic, and stores or retrieves data like user information, milk availability, etc. The User Management Service provides services for user registration, login, and authentication for both farmers and consumers. The Milk Inventory Service provides a platform for farmers to update their daily milk availability, both morning and evening

The Order Management Service is responsible for handling orders from consumers and checking for milk availability before processing the order. The Subscription Service allows for recurring orders for milk delivery, helping consumers schedule their orders for daily milk delivery. The Delivery Management Service oversees the entire delivery process, from allocating delivery partners to tracking the delivery status. Another backend service, Excess Milk Management Service, helps in reducing milk wastage by providing options for value-added products like paneer, butter, or ghee when milk is not sold.

### **2.4. DATABASE SCHEMA :**

| Table Name        | Description                                              |
|-------------------|----------------------------------------------------------|
| Users             | Stores farmer and consumer account information           |
| Farms             | Contains farm location, production capacity, and pricing |
| Milk_Availability | Records daily milk production and availability           |
| Orders            | Stores consumer order details                            |
| Deliveries        | Tracks delivery assignments and status                   |
| Products          | Lists value-added dairy products                         |
| Transactions      | Stores payment and order records                         |

## 2.5. APIs :

The system utilizes RESTful APIs for the purpose of enabling communication between the frontend user interface and the backend services. This facilitates users in performing operations like registration, updation of milk availability, order placement, etc. It helps in the smooth flow of data between the user interface and the backend system for efficient operation.

| API Endpoint    | Function                           |
|-----------------|------------------------------------|
| /register       | Register farmer or consumer        |
| /add_milk       | Update daily milk availability     |
| /farms          | Retrieve nearby farm listings      |
| /order          | Place milk order                   |
| /orders         | Retrieve order history             |
| /inventory      | View available milk stock          |
| /value-products | Display value-added dairy products |

## 2.6. SECURITY MECHANISMS :

In order to make the system reliable and secure, a number of security features are integrated into the system. User authentication is achieved through a secure login system where passwords are stored using hashing algorithms provided by Werkzeug password hashing. Data encryption is used to secure the data being transferred between the client and server. Role-based access control is integrated into the system to provide security. Farmers and consumers have different access levels. Input validation is done to prevent malicious input to the system to secure the database. Session management is used to provide secure sessions to the users.

## **2.7. IMPLEMENTATION :**

### **2.7.1. ALGORITHM :**

1. Input: Location of consumer
2. Retrieve all farm locations from database
3. Compute the distance between consumer and farm locations
4. Filter farm locations within delivery radius
5. Sort farm locations based on distance and availability
6. Display available farm locations to consumer

### **2.7.2. MATHEMATICAL FORMULA :**

#### **DISTANCE CALCULATION FOR HYPERLOCAL FARM MATCHING :**

$$d = 2R \times \arcsin \left( \sqrt{\sin^2 \left( \frac{\phi_2 - \phi_1}{2} \right) + \cos(\phi_1) \cos(\phi_2) \sin^2 \left( \frac{\lambda_2 - \lambda_1}{2} \right)} \right)$$

- $d$  = Distance between the farmer and the consumer
- $R$  = Radius of the Earth (approximately 6371 km)
- $\phi_1$  = Latitude of the consumer
- $\phi_2$  = Latitude of the farm
- $\lambda_1$  = Longitude of the consumer
- $\lambda_2$  = Longitude of the farm

The formula will compute the great-circle distance between the farmer's current position and the consumer's current position. The system will then use this distance to filter and show farms within a 20–30 km radius, ensuring that fresh milk is provided to the consumer in a fast and efficient manner.

#### **CALCULATION FOR DELIVERY TIME ESTIMATION :**

$$T = \frac{D}{V}$$

- $T$  = Delivery time
- $D$  = Distance between the farm and the consumer
- $V$  = Average delivery speed

The formula above is used to compute an approximate time for the milk to be delivered to the consumer. The formula works by dividing the distance by the average delivery speed to determine an approximate time for delivery to the consumer.

## **RESULTS AND DISCUSSIONS :**

The proposed Hyperlocal Direct Milk Supply Platform was developed as a functional prototype to assess its efficiency in enhancing the conventional milk supply system. The system was tested with key features such as farmer registration, milk availability updates, order processing, and delivery management. The results show that the proposed system can facilitate the interaction between farmers and consumers within a geographical radius. During the implementation of the system, farmers were able to add their details and update their daily milk availability, including morning and evening availability. The system reflected this information on the consumer interface, enabling consumers to order milk according to availability. The system facilitated order processing only when the quantity of milk was available. Thus, it ensured consistency in inventory management. The hyperlocal farm matching feature of the system was crucial in enhancing delivery efficiency. The system helped identify farms within a radius of 20-30 km. This ensured that the distance for milk delivery was minimal. Hence, it helped in maintaining the freshness of milk products. Freshness is a critical factor for consumers. The delivery management feature of the system helped in proper delivery order management. The subscription feature, as introduced in the system, would provide a regular demand for the farmers. The customers would be able to subscribe to the milk delivery on a daily basis or as scheduled, and this would help the farmers predict the demand and produce milk accordingly. Another significant impact of the system was the management of excess milk supply. The system would have a feature that would help the farmers identify the unsold milk and would suggest the production of value-added products like paneer, butter, ghee, and cheese from the milk. This would not only reduce the wastage of milk but would also provide the farmers with a new source of income. From the point of view of the performance of the system, the inclusion of the backend services would enable the effective functioning of the system. The utilization of RESTful APIs would allow the effective communication between the frontend and the backend, thus enabling faster responses and the exchange of data. However, some challenges were noted during the testing process. The accuracy of the hyperlocal matching relies on accurate location information, and any error in this will affect the accuracy of the farm recommendations. The system, although tested, is still in its prototype state and has not been tested in a large-scale real-world scenario. The system will be subjected to high user load, latency, and timely delivery issues, which will affect its performance in a live scenario.

The results show that the system, as proposed, will effectively address the issues in the traditional milk supply chain, including the lack of transparency, efficiency, and wastage. The above discussion indicates that although the system performs well at the prototype level, there is still room for improvements and live scenario deployments.



## **ADVANTAGES AND LIMITATIONS :**

### **ADVANTAGES :**

- Removal of intermediaries would enable a higher profit margin for the farmers.
- Fresh and unprocessed milk is being delivered.
- Delivery is hyperlocal, reducing the time taken in transportation and maintaining the quality.
- The subscription feature would enable a regular and predictable income stream for the farmers.
- Excess milk is being utilized, reducing wastage and enabling the sale of value-added products.
- Transparency is being provided to the end consumer about the source and quality of milk.

### **LIMITATIONS :**

- Internet connectivity is a prerequisite for the functioning of the entire system, which might be a problem in rural areas.
- Location detection is critical in the entire process of effective farm matching.
- The adoption of the entire system might be a problem in the initial stages due to a lack of digital knowledge among the farming community.
- Delivery might become a problem as the user base increases.
- The entire system is a prototype and has not undergone real-time testing at a large scale.

### **FUTURE WORK :**

The suggested system may be further enhanced and enriched with new and advanced features and scalability improvements in the future. The new features may include the incorporation of AI models that aid in the prediction of milk demand, thus enabling the farmer to plan the production of milk in a more efficient manner. Further, smart route optimization may be included in the system to improve the efficiency of delivery and reduce the costs of transportation. The suggested system may be further enhanced and enriched by incorporating milk quality testing using IoT devices. The suggested system may be further expanded and enriched to include mobile applications, thus making it easily accessible in rural areas. Further, the suggested system may be enhanced and enriched by incorporating online payment systems and wallets, thus making the transaction process more convenient. The suggested system may be further enhanced and enriched by making it a nationwide solution, thus enabling the inclusion of multiple regions and cooperative societies.

## **CONCLUSION :**

The Hyperlocal Direct Milk Supply Network that is proposed in this project is able to overcome the major issues that exist in the conventional milk supply chain. The issues faced in the conventional milk supply chain include transparency issues, lower returns to farmers because of middlemen, and milk wastage. The proposed system is able to overcome these issues through the implementation of the digital platform that can connect farmers to consumers and make the milk supply process more efficient and transparent. The developed prototype is able to show how technology can make the milk supply management process easier. The system can help farmers update their daily milk availability and allow consumers to order milk according to the availability. The implementation of the hyperlocal matching system can help identify the farms within a specific radius and make the transportation faster to ensure the quality of milk. One of the most notable aspects of this system is the integration of the subscription-based delivery system, which ensures a stable flow of income for the farmers. Moreover, the system also allows the farmers to plan their production more efficiently. Additionally, the integration of the excess milk management system has also played a vital role, as it helps the farmers avoid the wastage of milk. It does so by suggesting the production of other dairy products, such as paneer, butter, ghee, and cheese, from the unsold milk. The system has also ensured the transparency of the data, as the consumers can easily access the details of the milk, including the source, availability, and pricing. It has helped build trust among the consumers, as they are assured of receiving fresh, unprocessed milk directly from the farms. In the context of the system, the structured system, RESTful APIs, and backend services have ensured the smooth operation of the system. Although this is a prototype implementation, the results show that the proposed method is both possible and successful in enhancing the local dairy value chains. Therefore, if the system is improved to make it scalable and to incorporate advanced technologies such as AI and IoT-based milk quality analysis, it can be scaled up to become a large-scale system.

In conclusion, the Hyperlocal Direct Milk Supply Platform is a sustainable solution that is technology-based and is beneficial to both farmers and consumers. It enhances farmers' income, ensures fresh milk delivery to consumers, minimizes milk loss, and encourages transparency in the dairy value chain. The project shows the possibility of digital platforms in transforming traditional agricultural value chains into modern and user-centric systems.

